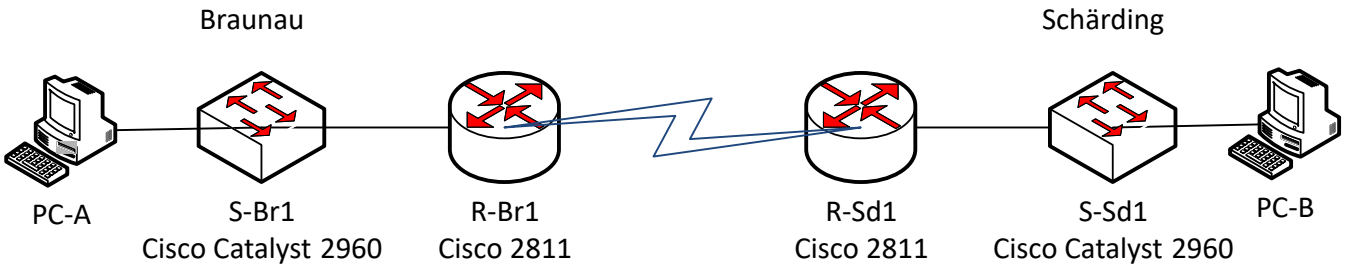
SYT-D Laborübungen Übung 03			
Klasse: 4AHITS	**Verfasser/in:** Denis Ermurachi	**Übung am:** 05.10.2026	**Abgabe am:** 11.10.2026
Übung: Standortvernetzung von 2 Standorten - IPv6			

Topologie



IPv4 Adresstabelle:

Device	Interface	IPv4 Address / Präfix	Default Gateway
R-Br1	FE0/0	192.168.1.1 /24	
	FE0/1	10.0.0.1/30	
R-Sd1	FE0/0	192.168.2.1 /24	
	FE0/1	10.0.0.2/30	
S-Br1	VLAN 1	192.168.1.2 /24	
S-Sd1	VLAN 1	192.168.2.2 /24	
PC-A		192.168.1.111/24	192.168.1.1
PC-B		192.168.2.111/24	192.168.2.1

IPv6 Adresstabelle:

IPv6-Präfix:

2001:db8:acad::/48

Device	Interface	IPv6 Address / Präfix	Default Gateway
R-Br1	FE0/0	2001:db8:acad:1::100/64	
	FE0/1	2001:db8:acad:2::1/64	
R-Sd1	FE0/0	2001:db8:acad:3::100/64	
	FE0/1	2001:db8:acad:2::2/64	
S-Br1	VLAN 1	2001:db8:acad:1::10/64	
S-Sd1	VLAN 1	2001:db8:acad:3::10/64	
PC-A		2001:db8:acad:1::1/64	2001:db8:acad:1::100/64
PC-B		2001:db8:acad:3::1/64	2001:db8:acad:3::100/64

Ziele

Teil 1: Einrichten der Topologie für eine Standortvernetzung mit 2 Standorten

Background / Scenario

Dies ist eine umfassende Übung, um IOS-Befehle zu überprüfen, die für eine Standortvernetzung von 2 Standorten mit IPv6 über eine WAN-Verbindung nötig sind. In dieser Übung erweitern sie die Konfiguration aus Übung 2 wie im Topologie-Diagramm gezeigt.
Anschließend konfigurieren Sie die Geräte entsprechend der selbst erstellten Adressierungstabelle bzw. Ihrer definierten Erweiterungen. Nachdem die Konfigurationen gespeichert wurden, überprüfen Sie Ihre Konfigurationen, indem Sie die Netzwerkkonnektivität testen.

Schritt 1: Konfiguration aller Geräte

Passwörter:

1. **Console- und Line vty-Passwort: cisco**
2. **Enable-Secret: class**

Welche statischen Routen werden an auf welchen Geräten benötigt?

Schritt 2:
Erweitern sie die bestehende IPv4 Konfiguration aus der Übung 02 um einen IPv6 Dual Stack. Entwickle dazu ein selbst gewähltes IPv6-Adresskonzept. Der ISP stellt dazu das Routing Präfix 2001:db8:acad: : /48 zur weiteren Verwendung zur Verfügung. Als Präfix-Länge für die Subnetze wird ein /64 Präfix empfohlen.

Schritt 3: Überprüfen Sie die End-to-End-Konnektivität.

a. Ping von PC-A zu S1/R1 – R2/S2 und PC-B
Alle Pings müssen erfolgreich sein

Schritt 5 Sichern Sie die Konfigurationen auf den Routern und fügen Sie die Konfigurationen in Ihrem Laborbericht ein.

Arbeitsbericht

- Name: Denis Ermurachi
- Datum: 05.10.2026
- Laborübung: Übung_03-Standortvernetzung_IPv6
- Fach: SYTD
- Klasse: 4AHITS

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1. Statische Routen:

   Am Router R-Br1 nu R-Sd1 werden folgende statische Routen benötigt, um die Kommunikation zwischen den beiden Standorten zu ermöglichen.

   R-Br1:

   ```
   ipv6 route 2001:DB8:ACAD:3::/64 2001:DB8:ACAD:2::2
   ```

   R-Sd1:

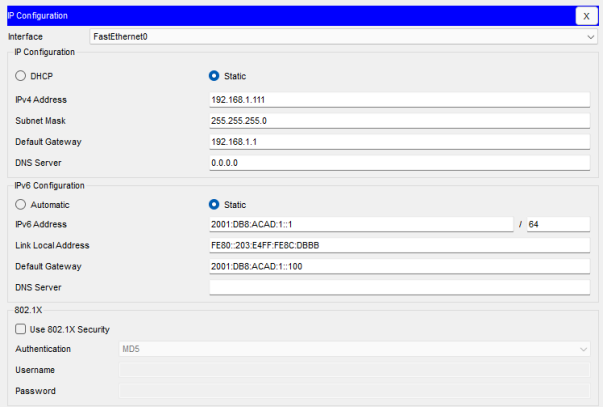
   ```
   ipv6 route 2001:DB8:ACAD:1::/64 2001:DB8:ACAD:2::1
   ```

---

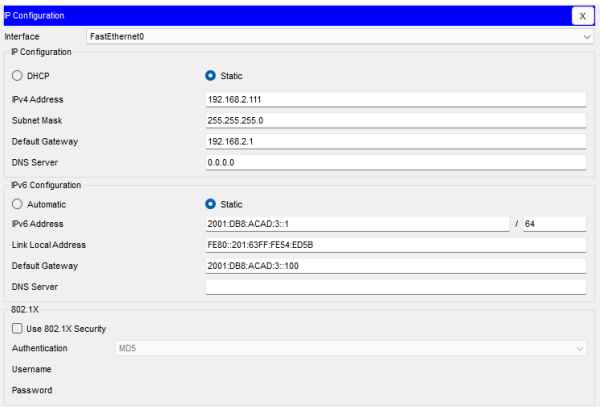
2. Konfigurationen:

   PCs

   PC-A

   PC-B

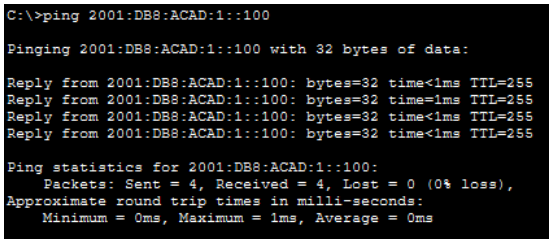
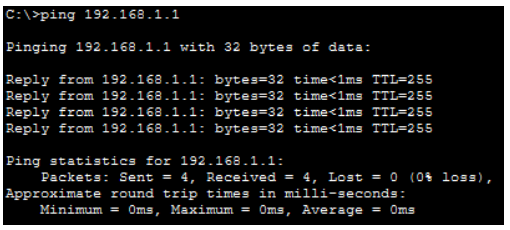
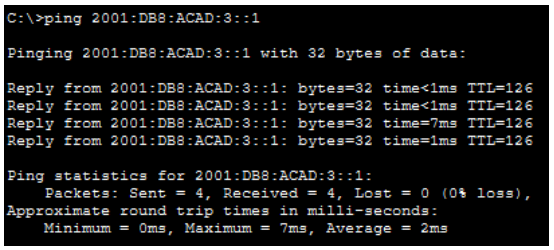
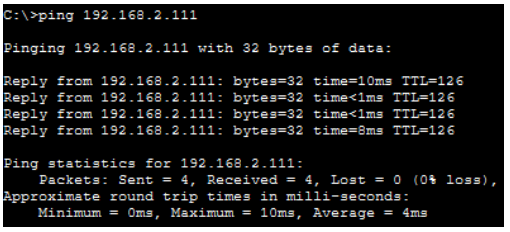
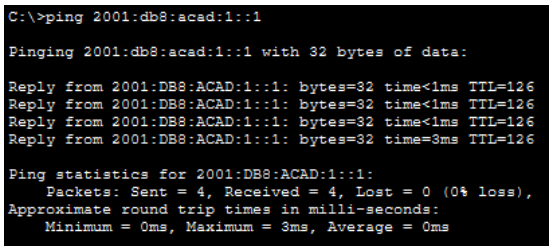
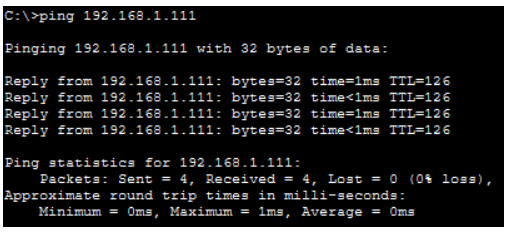
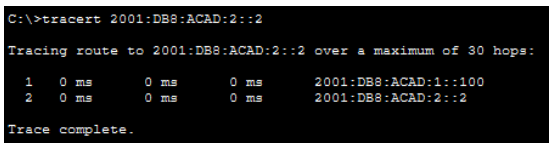
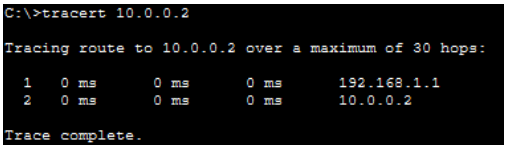
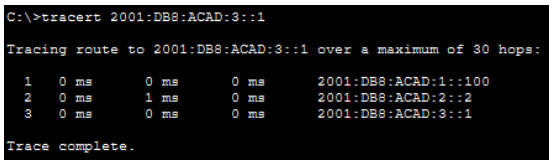
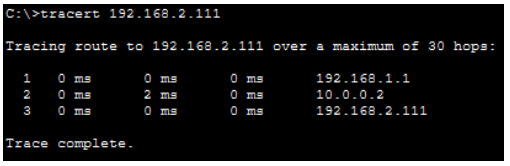
   Switches

	S-Br1	S-Sd1
Vlan1 Ipv6	Vlan1 [up/up] FE80::205:5EFF:FE57:92A1 2001:DB8:ACAD:1::10	Vlan1 [up/up] FE80::202:4AFF:FE01:E864 2001:DB8:ACAD:3::10
Vlan1 Interface	S-Br1#show int vlan1 Vlan1 is up, line protocol is up Hardware is CPU Interface, address is 0005.5e57.92a1 (bia 0005.5e57.92a1) Internet address is 192.168.1.2/24 MTU 1500 bytes, BW 100000 Kbit, DLY 1000000 usec, reliability 255/255, txload 1/255, rxload 1/255 Encapsulation ARPA, loopback not set ARP type: ARPA, ARP Timeout 04:00:00 Last input 21:40:21, output never, output hang never Last clearing of "show interface" counters never Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0 Queueing strategy: fifo Output queue: 0/40 (size/max) 5 minute input rate 0 bits/sec, 0 packets/sec 5 minute output rate 0 bits/sec, 0 packets/sec 1682 packets input, 530955 bytes, 0 no buffer Received 0 broadcasts (0 IP multicast) 0 runts, 0 giants, 0 throttles 0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored 563859 packets output, 0 bytes, 0 underruns 0 output errors, 23 interface resets 0 output buffer failures, 0 output buffers swapped out	S-Sd1#show interfaces Vlan1 Vlan1 is up, line protocol is up Hardware is CPU Interface, address is 0002.4a01.e864 (bia 0002.4a01.e864) Internet address is 192.168.2.2/24 MTU 1500 bytes, BW 100000 Kbit, DLY 1000000 usec, reliability 255/255, txload 1/255, rxload 1/255 Encapsulation ARPA, loopback not set ARP type: ARPA, ARP Timeout 04:00:00 Last input 21:40:21, output never, output hang never Last clearing of "show interface" counters never Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0 Queueing strategy: fifo Output queue: 0/40 (size/max) 5 minute input rate 0 bits/sec, 0 packets/sec 5 minute output rate 0 bits/sec, 0 packets/sec 1682 packets input, 530955 bytes, 0 no buffer Received 0 broadcasts (0 IP multicast) 0 runts, 0 giants, 0 throttles 0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored 563859 packets output, 0 bytes, 0 underruns 0 output errors, 23 interface resets 0 output buffer failures, 0 output buffers swapped out

Routers

	R-Br1	R-Sd1
Ipv6 Adressen	R-Br1#show ipv6 int brief FastEthernet0/0 [up/up] FE80::201:63FF:FE84:9401 2001:DB8:ACAD:1::100 FastEthernet0/1 [up/up] FE80::201:63FF:FE84:9402 2001:DB8:ACAD:2::1 Vlan1 [administratively down/down] unassigned R-Br1#	R-Sd1#show ipv6 int brief FastEthernet0/0 [up/up] FE80::209:7CFF:FE1E:6032 2001:DB8:ACAD:3::100 FastEthernet0/1 [up/up] FE80::250:FFF:FE0D:4B81 2001:DB8:ACAD:2::2 Vlan1 [administratively down/down] unassigned R-Sd1#
Interface FE0/0	R-Br1#show int f0/0 FastEthernet0/0 is up, line protocol is up (connected) Hardware is Lance, address is 0001.6384.9401 (bia 0001.6384.9401) Internet address is 192.168.1.1/24 MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec, reliability 255/255, txload 1/255, rxload 1/255 Encapsulation ARPA, loopback not set Full Duplex, 100Mbps, link type is auto, media type is RJ45 ARP type: ARPA, ARP Timeout 04:00:00, Last input 00:00:08, output 00:00:05, output hang never Last clearing of "show interface" counters never Input queue: 0/75/0 (size/max/drops); Total output drops: 0 Queueing strategy: fifo Output queue :0/40 (size/max) 5 minute input rate 0 bits/sec, 0 packets/sec 5 minute output rate 4 bits/sec, 0 packets/sec 334 packets input, 22512 bytes, 0 no buffer Received 0 broadcasts, 0 runts, 0 giants, 0 throttles 0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort 0 input packets with dribble condition detected 233 packets output, 17932 bytes, 0 underruns 0 output errors, 0 collisions, 1 interface resets 0 babbles, 0 late collision, 0 deferred 0 lost carrier, 0 no carrier 0 output buffer failures, 0 output buffers swapped out	R-Sd1#show int f0/0 FastEthernet0/0 is up, line protocol is up (connected) Hardware is Lance, address is 0009.7c1e.6032 (bia 0009.7c1e.6032) Internet address is 192.168.2.1/24 MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec, reliability 255/255, txload 1/255, rxload 1/255 Encapsulation ARPA, loopback not set Full Duplex, 100Mbps, link type is auto, media type is RJ45 ARP type: ARPA, ARP Timeout 04:00:00, Last input 00:00:08, output 00:00:05, output hang never Last clearing of "show interface" counters never Input queue: 0/75/0 (size/max/drops); Total output drops: 0 Queueing strategy: fifo Output queue :0/40 (size/max) 5 minute input rate 0 bits/sec, 0 packets/sec 5 minute output rate 4 bits/sec, 0 packets/sec 84 packets input, 5772 bytes, 0 no buffer Received 0 broadcasts, 0 runts, 0 giants, 0 throttles 0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort 0 input packets with dribble condition detected 122 packets output, 9280 bytes, 0 underruns 0 output errors, 0 collisions, 1 interface resets 0 babbles, 0 late collision, 0 deferred 0 lost carrier, 0 no carrier 0 output buffer failures, 0 output buffers swapped out
Interface FE0/1	R-Br1#show int f0/1 FastEthernet0/1 is up, line protocol is up (connected) Hardware is Lance, address is 0001.6384.9402 (bia 0001.6384.9402) Internet address is 10.0.0.1/30 MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec, reliability 255/255, txload 1/255, rxload 1/255 Encapsulation ARPA, loopback not set Full Duplex, 100Mbps, link type is auto, media type is RJ45 ARP type: ARPA, ARP Timeout 04:00:00, Last input 00:00:08, output 00:00:05, output hang never Last clearing of "show interface" counters never Input queue: 0/75/0 (size/max/drops); Total output drops: 0 Queueing strategy: fifo Output queue :0/40 (size/max) 5 minute input rate 4 bits/sec, 0 packets/sec 5 minute output rate 4 bits/sec, 0 packets/sec 119 packets input, 7864 bytes, 0 no buffer Received 0 broadcasts, 0 runts, 0 giants, 0 throttles 0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort 0 input packets with dribble condition detected 187 packets output, 12864 bytes, 0 underruns 0 output errors, 0 collisions, 1 interface resets 0 babbles, 0 late collision, 0 deferred 0 lost carrier, 0 no carrier 0 output buffer failures, 0 output buffers swapped out	R-Sd1#show int f0/1 FastEthernet0/1 is up, line protocol is up (connected) Hardware is Lance, address is 0050.0f0d.4b81 (bia 0050.0f0d.4b81) Internet address is 10.0.0.2/30 MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec, reliability 255/255, txload 1/255, rxload 1/255 Encapsulation ARPA, loopback not set Full Duplex, 100Mbps, link type is auto, media type is RJ45 ARP type: ARPA, ARP Timeout 04:00:00, Last input 00:00:08, output 00:00:05, output hang never Last clearing of "show interface" counters never Input queue: 0/75/0 (size/max/drops); Total output drops: 0 Queueing strategy: fifo Output queue :0/40 (size/max) 5 minute input rate 4 bits/sec, 0 packets/sec 5 minute output rate 4 bits/sec, 0 packets/sec 197 packets input, 13668 bytes, 0 no buffer Received 8 broadcasts, 0 runts, 0 giants, 0 throttles 0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort 0 input packets with dribble condition detected 120 packets output, 7944 bytes, 0 underruns 0 output errors, 0 collisions, 1 interface resets 0 babbles, 0 late collision, 0 deferred 0 lost carrier, 0 no carrier 0 output buffer failures, 0 output buffers swapped out
Routes	R-Br1#show ipv6 route IPv6 Routing Table - 6 entries Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP U - Per-user Static route, M - MIPv6 I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2 ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2 D - EIGRP, EX - EIGRP external C 2001:DB8:ACAD:1::/64 [0/0] via FastEthernet0/0, directly connected L 2001:DB8:ACAD:1::100/128 [0/0] via FastEthernet0/0, receive C 2001:DB8:ACAD:2::/64 [0/0] via FastEthernet0/1, directly connected L 2001:DB8:ACAD:2::1/128 [0/0] via FastEthernet0/1, receive S 2001:DB8:ACAD:3::/64 [1/0] via 2001:DB8:ACAD:2::2 L FF00::/8 [0/0] via Null0, receive	R-Sd1#show ipv6 route IPv6 Routing Table - 6 entries Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP U - Per-user Static route, M - MIPv6 I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2 ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2 D - EIGRP, EX - EIGRP external S 2001:DB8:ACAD:1::/64 [1/0] via 2001:DB8:ACAD:2::1 C 2001:DB8:ACAD:2::/64 [0/0] via FastEthernet0/1, directly connected L 2001:DB8:ACAD:2::2/128 [0/0] via FastEthernet0/1, receive C 2001:DB8:ACAD:3::/64 [0/0] via FastEthernet0/0, directly connected L 2001:DB8:ACAD:3::100/128 [0/0] via FastEthernet0/0, receive L FF00::/8 [0/0] via Null0, receive

3. Konnektivitätstest:

von - zu	ipv6	ipv4
ping PC-A ↓ S-Br1		
ping PC-A ↓ PC-B		
ping PC-B ↓ PC-A		
tracert PC-A ↓ S-Sd1		
tracert PC-A ↓ PC-B		
tracert PC-B ↓ PC-A	